2023 AMC 10B Problem 13 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 13. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 13

Problem:

What is the area of the region in the coordinate plane defined by

$$||x|-1|+||y|-1| \leq 1?$$

Answer Choices:

A. 2

B. 8

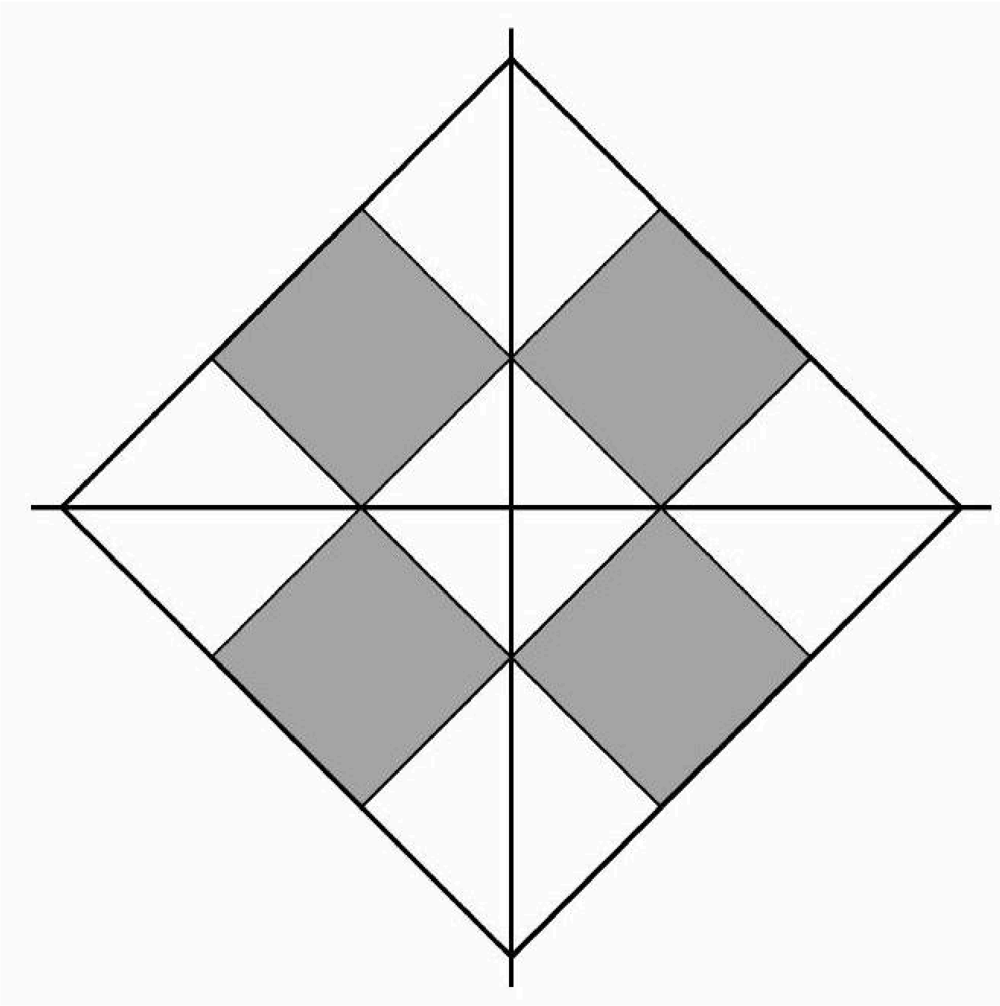
C. 4

D. 15

E. 12

Solution:

Replacing x by $-x$ and y by $-y$ does not change the inequality, so the region is symmetric with respect to both coordinate axes. In the first quadrant the given inequality is equivalent to $|x-1|+|y-1| \leq 1$, and its graph is the square bounded by the lines $x+y=1, x+y=3, x-y=-1$, and $x-y=1$ together with its interior. The area of this square is 2 . Therefore the area of the entire region is $2.4=(\mathrm{B})\boxed{8}$. (The region consists of the 4 shaded squares in the figure below.)



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